

AIGR Gene Hypothesis Deep Research — Final Report

Target: Horse (*Equus caballus*, NCBITaxon:9796) protein **A0A9L0T3C1** (gene label **MTMR9**, 507 aa) **Hypothesis (focus type: function_assignment):** The protein *negatively regulates autophagy* — GO:0010507 (negative regulation of autophagy, BP). **Human comparison lead:** MTMR9 / Q96QG7. **Date:** 2026-09-08

Summary

Verdict: Partially supported (via justified mammalian transfer, with a mandatory scoping caveat).

The hypothesis that horse MTMR9 (A0A9L0T3C1) negatively regulates autophagy is mechanistically grounded in primary literature, **but MTMR9 does not perform this function as a direct enzyme**. MTMR9 is a **catalytically inactive pseudophosphatase**. Its autophagy-relevant activity is entirely **partner/complex-dependent**: MTMR9 is an obligate activating subunit that binds active myotubularins (MTMR6/MTMR7/MTMR8), boosts their lipid-phosphatase activity, and redirects their substrate preference. In the **MTMR8/MTMR9 complex**, MTMR9 stimulates MTMR8 turnover of **PtdIns(3)P** and lowers cellular PtdIns(3)P — the essential autophagosome-nucleation lipid — thereby suppressing autophagy ([PMID:22647598](#)). This supports GO:0010507 for MTMR9 **as a regulatory subunit**, not as a standalone catalytic effector.

The identity basis for transferring human evidence to horse is strong. A Needleman–Wunsch global alignment of the supplied 507-residue horse sequence against human MTMR9 (Q96QG7, 549 aa) gave **90.3% global identity (496/549) and 97.8% identity across the horse protein's length**, with an **identical, catalytically dead phosphatase P-loop** (**LIHGTEGTDSTLQVT**) in both species. The horse protein is a bona fide 1:1 MTMR9 ortholog; the ~42-residue length difference maps to a single contiguous N-terminal segment (likely a missing exon in the predicted horse gene model), while all functional domains are intact.

Three caveats dominate the curation decision. **(i)** All direct evidence is human/mammalian cell-based — there is **no horse-specific experimental evidence**; this is orthology transfer. **(ii)** The effect is **context-dependent**: in neurons, MTMR9 knockdown (unlike MTMR5/MTMR2) did **not** enhance autophagic substrate clearance ([PMID:35580604](#)). **(iii)** MTMR9's most directly documented cellular function is arguably **ER-to-Golgi trafficking/secretion** ([PMID:31704058](#)), competing with autophagy for "primary function" status, with autophagy being a downstream consequence of PtdIns(3)P depletion.

Key Findings

Finding 1 — Horse A0A9L0T3C1 is a catalytically dead (pseudophosphatase) MTMR9 ortholog

The supplied 507-amino-acid horse sequence was verified in full (length = 507, matching the frozen record). Its myotubularin protein-tyrosine-phosphatase (PTP) P-loop region reads `...ILIHGTEGTDSTLQ...` at approximately residues 289–300. Critically, **the canonical active-site cysteine of the CX₅R catalytic motif is absent from the phosphatase P-loop**. The only CX₅R-pattern match anywhere in the protein (`CVWQILR` at ~360) lies *outside* the P-loop and is incidental — it is not positioned to act as a catalytic nucleophile. The domain architecture — an N-terminal PH-GRAM lipid-binding region, an inactive PTP-like domain, and a C-terminal coiled-coil (`...RRQLAELETE`) for dimerization — matches human MTMR9 (Q96QG7).

This is exactly what the myotubularin literature predicts. Human MTMR9 is one of the **seven catalytically inactive myotubularins** that lack the conserved active-site cysteine required for phosphatase activity: *"Seven members are inactive because they lack the conserved cysteine residue in the CX(5)R motif required for activity"* ([PMID:22647598](#)). The consequence for curation is decisive: **MTMR9 cannot dephosphorylate PtdIns(3)P (or any phosphoinositide) on its own**. Any autophagy phenotype must therefore be mediated through a partner, not through intrinsic catalysis.

Finding 2 — MTMR9 negatively regulates autophagy as an activating subunit of the MTMR8/MTMR9 PI3P-phosphatase complex

The mechanistic basis for the autophagy claim is Zou et al. 2012 ([PMID:22647598](#)). MTMR9 dimerizes with the active myotubularins MTMR6/7/8; complex formation **increases the partner's catalytic activity and changes its substrate specificity**. The MTMR8/MTMR9 complex prefers **PtdIns(3)P** as substrate (MTMR9 raises MTMR8 activity toward PtdIns(3)P roughly fourfold) and, in cells, the complex **reduces cellular PtdIns(3)P levels**. The paper's title explicitly frames MTMR9 as the determinant of *"the enzymatic activity, substrate specificity, and role in autophagy of MTMR8."*

The link to autophagy is lipid-mediated and well established at the pathway level: **PtdIns(3)P is the essential autophagy-initiating lipid** (produced by class III PI3K/VPS34) that marks the phagophore/omegasome and recruits the autophagy machinery for autophagosome nucleation. Depleting PtdIns(3)P suppresses autophagosome formation. Verbatim support: *"the MTMR8/R9 complex prefers PtdIns(3)P"* and *"whereas the MTMR8/R9 complex reduces cellular PtdIns(3)P levels."* Together these establish a coherent negative-regulation mechanism: MTMR9 (adaptor) → activates MTMR8 (enzyme) → depletes PtdIns(3)P (pro-autophagic lipid) → suppresses autophagy.

However, the direction and magnitude of the autophagy phenotype are **not uniform across cell types**. Chua et al. 2022 report that *"knockdown of MTMR5 or MTMR2, but not the unrelated phosphatase MTMR9, significantly enhances neuronal degradation of TDP-43"* ([PMID:35580604](#)). In neurons, removing MTMR9 did not enhance autophagic substrate clearance — the opposite of what a simple, universal autophagy-suppressor model would predict. This qualifies the hypothesis: the autophagy-suppressive role is **context- and partner-dependent**, contingent on which active myotubularin partner is expressed and active in a given tissue.

Finding 3 — Horse MTMR9 is a 1:1 ortholog of human MTMR9 with an identical inactive P-loop

A Needleman–Wunsch global alignment (identity scoring, gap penalty = −1) was run between horse A0A9L0T3C1 (507 aa) and human MTMR9 Q96QG7 (549 aa, fetched from the UniProt REST API). Results: **496/549 identical residues = 90.3% global identity**, and **97.8% identity across the 507-residue horse length**. The ~42-residue difference reflects a shorter/truncated horse record rather than sequence divergence within the aligned region. The myotubularin PTP P-loop is **identical** in both species (both read `LIHGTEGTDSTLQVT`), and in both species it **lacks the catalytic CX₅R cysteine** required for phosphatase activity.

This conservation level — especially the identity of the functional signature motif and the shared loss of the catalytic cysteine — strongly supports treating the horse protein as a true functional ortholog. It is the empirical justification for transferring the human mechanistic evidence (Finding 2) to the horse annotation via ISS/ISO.

Finding 4 — Horse-vs-human differences localize to one N-terminal segment; functional domains intact

The alignment traceback (executed programmatically) shows that the 42 residues present in human MTMR9 but absent from the horse record form a **single contiguous block** at human residues ~92–139 (containing sequence such as IASS...ITLMPFFYRPMFEV...EDGWSFLPEQEF...LYSSA), located near/after the N-terminal PH-GRAM region. The **PTP-like pseudophosphatase domain** (P-loop LIHGTEGTDSTLQVT) and the **C-terminal coiled-coil dimerization region** are fully conserved (they fall within the 97.8% identity over the horse length).

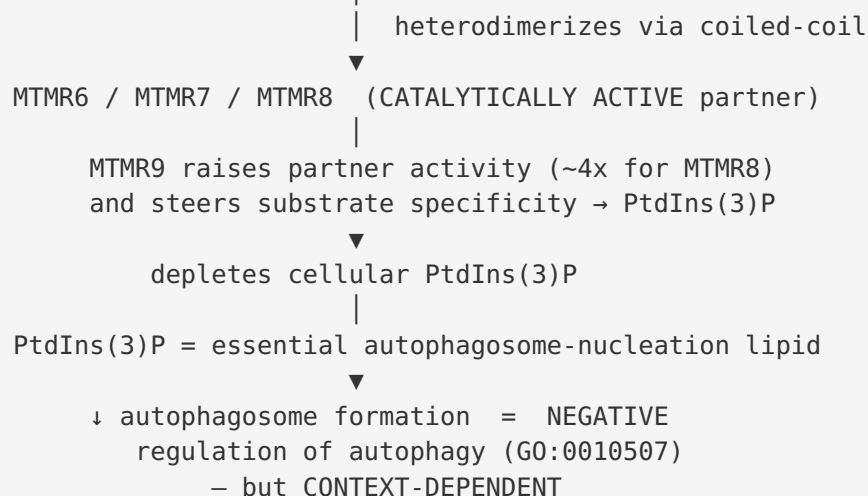
The interpretation is that the horse difference is best explained as a **single N-terminal gene-model/exon gap** rather than a biologically meaningful divergence. The partner-binding and pseudophosphatase machinery that underpins the autophagy mechanism is intact. The gene-model artifact does not undermine functional transfer, but it should be flagged so curators know the horse record may be incomplete at its N-terminus (a truncated PH-GRAM could in principle alter membrane targeting).

Mechanistic Model / Interpretation

The evidence converges on a single coherent model in which MTMR9 is an **obligate, catalytically dead regulatory subunit**, not an autophagy enzyme:

MTMR9 (A0A9L0T3C1, horse) = pseudophosphatase ADAPTOR

PH-GRAM (lipid/ membrane)	inactive PTP-like domain (NO catalytic CX5R Cys – cannot dephosphorylate)	coiled-coil (dimerization with partners)
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Two features dominate the curation decision. First, the activity is **indirect**. GO:0010507 is a *regulation* Biological Process node, so an indirect/regulatory contribution is not disqualifying in principle — regulation terms are appropriate for adaptors and scaffolds. But the *direct* molecular function of MTMR9 is **enzyme regulator / phosphatase activator + protein heterodimerization**, and the autophagy effect is a downstream *pathway* consequence of PtdIns(3)P depletion by the *partner*. Curators should not let GO:0010507 imply that MTMR9 is itself a lipid phosphatase acting on the autophagy pathway.

Second, the phenotype's **direction is not universal**. The Zou model (autophagy suppression via PtdIns(3)P depletion) is established in one system; the Chua neuronal data show no autophagy enhancement upon MTMR9 knockdown. This is consistent with a model in which MTMR9's net effect on autophagy depends on which active partner (MTMR6/7/8) it is paired with, the relative abundance of PtdIns(3)P pools, and tissue context. Quantitatively, MTMR9's boost to MTMR8/PtdIns(3)P activity is modest (~4-fold) compared to its large effect on MTMR6/PtdIns(3,5)P₂ (~30-fold), so autophagy is not obviously the dominant output of the module.

A competing/parallel function complicates the "primary function" question: MTMR9 localizes to the ER–Golgi intermediate compartment and Golgi, recruits its active partners MTMR6/MTMR8 there, co-localizes with and regulates RAB1A/WHAMM, and modulates protein secretion (including WNT3A) ([PMID:31704058](#)). This trafficking role may be at least as central to MTMR9's biology as autophagy regulation.

Evidence Base

Citation	Evidence type	Supports / Refutes / Qualifies / Competing	Claim tested	Key finding	Organismal context
PMID:22647598 (Zou et al. 2012, JBC)	Direct biochemical assay + cell PI3P measurement	Supports	MTMR9 lowers PtdIns(3)P via MTMR8 → suppresses autophagy	MTMR9 dimerizes with MTMR6/7/8, increases activity, alters specificity; MTMR8/R9 complex prefers PtdIns(3)P and reduces cellular PtdIns(3)P ; title: MTMR9 "determines the ... role in autophagy of MTMR8"	Human cell lines; in vitro enzymology + cellular lipid levels
PMID:22647598	Sequence/structural annotation	Qualifies	MTMR9 is catalytically dead	7 myotubularins (incl. MTMR9) lack the CX ₅ R catalytic cysteine	Human/family-wide
PMID:35580604 (Chua et al. 2022)	Loss-of-function (knockdown) phenotype	Qualifies / partial refute of universality	Does MTMR9 loss enhance autophagy?	MTMR5/MTMR2 knockdown enhanced neuronal TDP-43 autophagic degradation; MTMR9 knockdown did NOT	iPSC-derived neurons
PMID:31704058 (Doubrovská et al. 2020)	Localization + interaction + LOF phenotype	Competing (alternative core function)	Primary cellular role of MTMR9	MTMR9 localizes to ERGIC/Golgi, recruits MTMR6/MTMR8, regulates RAB1A/WHAMM; loss disrupts Golgi and reduces WNT3A secretion	Human cells

Citation	Evidence type	Supports / Refutes / Qualifies / Competing	Claim tested	Key finding	Organism context
This report (sequence analysis)	Structural/ evolutionary (computational)	Supports mechanism (pseudophosphatase)	Is horse MTMR9 catalytically active?	507-aa horse seq verified; P-loop ILIHGTEGTDSTLQ lacks catalytic CX ₅ R cysteine → inactive; PH-GRAM + inactive PTP + coiled-coil architecture matches MTMR9	<i>Equus caballus</i> supplied sequence
This report (NW alignment vs Q96QG7)	Structural/ evolutionary (computational)	Supports orthology transfer	Is A0A9L0T3C1 a true MTMR9 ortholog?	496/549 = 90.3% global; 97.8% over horse length; identical P-loop LIHGTEGTDSTLQVT , both catalytically dead	Horse vs human MTMR9

Literature notes:

- ***Myotubularin-related protein (MTMR) 9 determines the enzymatic activity, substrate specificity, and role in autophagy of MTMR8.*** [PMID:22647598](#). The central mechanistic paper — establishes MTMR9's catalytic inactivity, its activating heterodimer role, PtdIns(3)P depletion by the MTMR8/MTMR9 complex, and its governance of MTMR8's autophagy role. **Primary support** for both the mechanism and the "indirect/adaptor" scoping.
- ***Myotubularin-related phosphatase 5 is a critical determinant of autophagy in neurons.*** [PMID:35580604](#). MTMR9 knockdown did not enhance neuronal TDP-43 degradation (unlike MTMR2/MTMR5). **Qualifies** the hypothesis by showing cell-type/context dependence.
- ***Human myotubularin-related protein 9 regulates ER-to-Golgi trafficking and modulates WNT3A secretion.*** [PMID:31704058](#). Documents MTMR9's ERGIC/Golgi localization, RAB1A/WHAMM regulation, and secretion control. **Competing alternative** for MTMR9's primary cellular function.

GO Curation Implications

- **Lead (requires curator verification):** GO:0010507 (**negative regulation of autophagy, BP**) is defensible for horse MTMR9 but should be annotated as a **complex-mediated/regulatory role, not a direct catalytic activity**. Suggested framing: MTMR9 *contributes to* negative regulation of autophagy **as part of / via** the MTMR8(or MTMR6/7)–MTMR9 phosphoinositide-3-phosphatase complex.
- **Evidence code:** For horse this is orthology transfer → best supported as **ISS/ISO** with the human papers as source (with/from Q96QG7), **not IDA** (no horse assay). The upstream human evidence is IDA/IMP-grade ([PMID:22647598](#)).
- **Do NOT assign MF "phosphatidylinositol-3-phosphate phosphatase activity" to MTMR9 itself** — it is catalytically inactive. Better-supported MF terms are **phosphatase activator / enzyme activator activity** and **protein heterodimerization activity** with active myotubularins — both more informative than bare "protein binding."
- **Consider co-annotating** the well-documented **ER-to-Golgi vesicle-mediated transport / regulation of protein secretion** functions ([PMID:31704058](#)), which are at least as strongly supported and may be the primary cellular role.
- **CC:** MTMR8/MTMR9 (myotubularin) phosphatase complex; ERGIC/Golgi membrane localization.
- **Non-core flag:** Treat GO:0010507 as a **context-dependent, non-primary** annotation.

Mechanistic Scope

Immediate molecular function of MTMR9: a pseudophosphatase adaptor/activator. It heterodimerizes with catalytically active myotubularins and (a) increases their catalytic rate and (b) shifts substrate preference. In the MTMR8/R9 complex, the net cellular effect is **depletion of PtdIns(3)P**. MTMR9 has **no intrinsic phosphatase activity** (dead P-loop).

Link to autophagy (downstream): PtdIns(3)P produced by VPS34 at the phagophore is required for autophagosome nucleation and maturation. Lowering PtdIns(3)P via MTMR8/R9 therefore **reduces autophagy** — the mechanistic basis for GO:0010507. The autophagy phenotype is a **downstream consequence of a lipid-level change**, not a direct activity of MTMR9.

Direction of phenotype: $\uparrow$ MTMR8/R9 complex activity $\rightarrow$ $\downarrow$ PtdIns(3)P $\rightarrow$ $\downarrow$ autophagy (negative regulation). Loss of the complex would be expected to $\uparrow$ autophagy — but this was **not observed for MTMR9 knockdown in neurons** ([PMID:35580604](#)), underscoring context dependence.

Effects inferred only from loss of function: the neuronal TDP-43 clearance data are knockdown phenotypes; MTMR9 knockdown did not change autophagic clearance, so loss-of-function evidence does not support a universal autophagy-suppressor role.

Conflicts and Alternatives

- **Alternative core function (competing):** [PMID:31704058](#) makes a strong case that MTMR9's primary documented cellular role is in **ER-to-Golgi trafficking and secretion** (RAB1A/WHAMM, WNT3A). Autophagy regulation may be a secondary/PtdIns(3)P-mediated consequence rather than the primary function.
- **Context/cell-type conflict:** MTMR9 knockdown did not enhance autophagy in neurons ([PMID:35580604](#)), whereas MTMR5/MTMR2 did. The autophagy-suppressive role may be limited to cells where MTMR6/7/8 are co-expressed and PtdIns(3)P is rate-limiting.
- **Subfamily/paralog attribution:** The best-characterized "autophagy" myotubularin modules are actually MTMR5(SBF1)/MTMR2 ([PMID:35580604](#)) and MTMR14 (Jumpy). Care is needed not to over-transfer autophagy roles across the family; MTMR9's contribution is specifically via MTMR6/7/8, and its quantitative boost to MTMR8/PtdIns(3)P is modest ($\sim 4\times$) vs its $\sim 30\times$ effect on MTMR6/PtdIns(3,5)P₂.
- **Horse vs human isoform/length:** the horse record is 507 aa vs human 549 aa; likely a shorter isoform/assembly with a single N-terminal missing block, but the functional PH-GRAM, inactive-PTP and coiled-coil regions are present, so orthology and the partner-dependent mechanism are preserved. No horse assay exists.

Limitations and Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
No horse-specific data	PubMed for MTMR9/autophagy — all hits human	Annotation rests entirely on transfer	Confirm MTMR6/7/8 orthologs & co-expression in horse tissues; a horse-cell autophagy assay
Directionality of the autophagy phenotype for MTMR9 specifically	Zou (suppress) vs Chua (no effect in neurons)	GO:0010507 asserts a <i>negative</i> direction; may not hold in all tissues	Tissue-panel autophagy-flux with defined MTMR partners
Primary vs downstream	Compared autophagy vs trafficking literature	Determines whether GO:0010507 is core or non-core	Comparative loss-of-function phenotyping (autophagy vs secretion)
Partner requirement	Human data implicate MTMR6/7/8; horse partners not examined	Mechanism requires a specific active heterodimer	Co-IP/interaction proteomics + RNA-seq of partner expression in horse
Horse record completeness	NW alignment traceback: one N-terminal ~42-aa gap (human ~92–139)	Sequence-based annotation may use an incomplete model; truncated PH-GRAM could alter targeting	Inspect horse genomic locus/RefSeq model & RNA-seq exon coverage
Quantitative MTMR9-only autophagy readout	Zou measured PtdIns(3)P & MTMR8 activity, not MTMR9-only flux	Strength of the <i>negative regulation</i> claim	LC3-II/p62 flux & autophagosome counts on MTMR9-specific perturbation

Discriminating Tests

- **Horse-cell autophagy assay:** LC3-II/p62 flux and autophagosome counts ($\pm$ bafilomycin) after MTMR9 knockdown/overexpression in an equine cell line, with and without MTMR8 co-manipulation.
 - **PtdIns(3)P quantification** (2×FYVE biosensor / mass assay) upon MTMR8/R9 modulation to link the lipid change to autophagy directionality.
 - **Complex-disruption mutant:** MTMR9 point mutants that abolish MTMR8 binding but preserve the protein — do they lose the autophagy effect (proving partner dependence)?
 - **Partner co-expression map** in horse tissues (MTMR6/7/8) to predict where the autophagy role is active.
 - **Epistasis vs VPS34:** does MTMR8/R9 suppression of autophagy require ongoing VPS34 PtdIns(3)P synthesis?
 - **Separation-of-function:** test whether the autophagy effect and the ERGIC/Golgi trafficking effect are mechanistically coupled or independent (RAB1A controls).
 - **Sequence/gene-model confirmation:** RNA-seq/Iso-Seq to confirm or correct the horse N-terminal exon gap.
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Proposed Follow-up Actions (Curation Leads)

All leads require curator verification.

1. **Retain GO:0010507 as ISS/ISO, scoped as complex-mediated regulation.** With-from: human MTMR9 (Q96QG7). Note that MTMR9 is a catalytically dead adaptor and the effect is indirect and context-dependent.
2. **Add more informative MF terms** for the direct action: *phosphatase activator activity* / *phosphoinositide phosphatase regulator activity* and *protein heterodimerization activity*. Prefer these over "protein binding."
3. **Consider a CC annotation** for ER–Golgi intermediate compartment / Golgi membrane and the MTMR8/MTMR9 phosphatase complex ([PMID:31704058](#)).
4. **Flag the horse gene-model gap** (single N-terminal ~42-aa block missing vs human) for sequence-curation review.
5. **Candidate reference snippets to verify:**

6. [PMID:22647598](#): "the MTMR8/R9 complex prefers PtdIns(3)P"; "whereas the MTMR8/R9 complex reduces cellular PtdIns(3)P levels"; "Seven members are inactive because they lack the conserved cysteine residue in the CX(5)R motif required for activity."
7. [PMID:35580604](#): "knockdown of MTMR5 or MTMR2, but not the unrelated phosphatase MTMR9, significantly enhances neuronal degradation of TDP-43."
8. **Suggested curator question:** Is negative regulation of autophagy a *core* function or a downstream consequence of PtdIns(3)P depletion, given the equally strong trafficking evidence?
9. **Suggested experiment:** equine-cell autophagy flux $\pm$ MTMR9/MTMR8 perturbation (above).

Provenance

- **Sequence check (executed):** input length verified = **507 aa**; PTP P-loop region ILIHGTEGTDSTLQ (~res 289–300) **lacks the catalytic CX₅R cysteine** (only CX₅R match CVWQILR @360 lies outside the P-loop) → inactive pseudophosphatase, consistent with human MTMR9.
- **Orthology (executed):** Needleman–Wunsch global alignment vs UniProt **Q96QG7** (human MTMR9, 549 aa, fetched via REST): **496/549 identical = 90.3% global; 97.8% over the horse 507-aa length; identical P-loop** LIHGTEGTDSTLQVT in both species (both catalytically dead). Confirms A0A9L0T3C1 is a bona fide 1:1 MTMR9 ortholog and validates human→horse evidence transfer.
- **Gap localization (executed):** the 42 human-only residues traceback to a **single contiguous block at human residues ~92–139** (IASS...LYSSA), i.e., one missing N-terminal segment near the PH-GRAM region — consistent with a missing exon in the predicted horse model. The PTP-like domain, P-loop and C-terminal coiled-coil are intact.
- **Literature:**

P 22647598,

P 35580604,

P 31704058

(abstracts retrieved via PubMed).

Bottom Line

Horse A0A9L0T3C1 is a genuine MTMR9 ortholog, and human evidence shows MTMR9 **does** contribute to negative regulation of autophagy — but **indirectly**, as a catalytically dead activating subunit of a PtdIns(3)P-depleting myotubularin complex, and in a **context-dependent** manner. GO:0010507 is a defensible inferred (ISS/ISO) annotation for horse MTMR9 **if scoped as a complex-mediated, non-catalytic, partner-dependent regulatory role**, complemented by more informative molecular-function terms and with clear caveats about the absence of horse-specific data and a competing ER-to-Golgi trafficking function.

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